1.05 Complex Numbers: Equations with Complex Coefficients

Worked Solutions

1 a)

$$\begin{aligned}&(z-(1+i))(z-(3+2i))\\&=z^2-(3+2i)z-(1+i)z+(1+i)(3+2i)\\&=z^2-(4+3i)z+1+5i\end{aligned}$$

1 b)

$$\begin{aligned}&(z-(-3+4i))(z-(1+5i))\\&=z^2-(1+5i)z-(-3+4i)z+(-3+4i)(1+5i)\\&=z^2+(2-9i)z-23-11i\end{aligned}$$

1 c)

$$\begin{aligned}&(z-(-2+2i))(z-(2-2i))\\&=z^2-(2-2i)z-(-2+2i)z+(-2+2i)(2-2i)\\&=z^2+8i\end{aligned}$$

2)

$$\begin{aligned}&(z-(1+i))(z-(-1+i))(z-(2+2i))\\&=(z^2-2iz-2)(z-(2+2i))\\&=z^3-(2+4i)z^2-(6-4i)z+(4+4i)\end{aligned}$$

3 a)

$$\begin{aligned}(1+9i)^3&=(1+9i)(1+9i)(1+9i)\\&=(1+9i)(-80+18i)\\&=-80-720i+18i-162\\&=-242-702i\end{aligned}$$

3 b)

$$\begin{aligned}&(1+9i)^3-(5+6i)(1+9i)^2+(35+27i)(1+9i)-58-30i\\&=-242-702i-(5+6i)(-80+18i)-208+342i-58-30i\\&=-242-702i+508+390i-266+312i\\&=0\end{aligned}$$

$\therefore z=1+9i$ is a solution.

3 c)

$$\begin{aligned}(z-2)(z-(1+9i)) &= z^2-(1+9i)z-2z+(2+18i)\\ &= z^2-(3+9i)z+(2+18i)\end{aligned}$$

So for some complex number k

$$(z^2-(3+9i)z+(2+18i))(z-k) = k^3-(5+6i)k^2+(35+27i)k-58-30i$$

Comparing constants:

$$-(2+18i)k = -58-30i$$

$$k = \frac{58+30i}{2+18i}\times\frac{2-18i}{2-18i} = \frac{656-984i}{328} = 2-3i$$

Alternative Method:

Sum of roots: $5+6i = (1+9i)+(2)+(a+bi)$

$\Rightarrow a = 2$, $b = -3$ so $2-3i$

4 a)

$$\begin{aligned}(1+2i)^2-(2+3i)(1+2i)-k &= 0\\ -3+4i+4-7i-k &= 0\\ \therefore k &= 1-3i\end{aligned}$$

4 b)

$$\begin{aligned}&(1-2i)^2-(2+3i)(1-2i)-(1-3i)\\ &= -3-4i-8+i-1+3i\\ &= -12\neq 0 \text{ so not a root.}\end{aligned}$$

5 a)

All the coefficients are not real, so must use factor theorem.

Since $2-i$ is a root, $\mathrm{f}(2-i)=0$

$$\begin{aligned}(2-i)^3-3i(2-i)^2+2(2-i)+k &= 0\\ &= (2-i)(5-4i)-3i(5-4i)+4-2i+k = 0\\ &= 2-11i-12-9i+4-2i+k = 0\\ &= -6-22i+k = 0\\ \therefore k &= 6+22i\end{aligned}$$

5 b)

$$\begin{aligned}&(2+i)^3-3i(2+i)^2+2(2+i)+6+22i\\ &= (2+i)(3+4i)-3i(3+4i)+4+2i+6+22i\\ &= 2+11i+12-9i+4+2i+6+22i\\ &= 24+26i\neq 0\end{aligned}$$

$\therefore$ by the factor theorem $2+i$ is not a factor and so $2+i$ is not a root of the equation $\mathrm{f}(z)=0$

6 a)

$$z^3 - i = (z+i)(z^2 - iz - 1)$$
$$\mathrm{f}(z) = z^2 - iz - 1$$

6 b)

$$(z+i)(z^2 - iz - 1) = 0$$

Consider the quadratic factor:

$$z = \frac{i \pm \sqrt{(-i)^2 + 4}}{2} = \frac{i \pm \sqrt{3}}{2} = \pm\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

So the solutions are $z = i$, $z = \frac{\sqrt{3}}{2} + \frac{1}{2}i$ and $z = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

7)

$$z = \frac{8 - 5i \pm \sqrt{(8-5i)^2 - 4(3-29i)}}{2}$$

Consider just the discriminant:

$$(8-5i)^2 - 4(-15i) = 39 - 80i - 12 + 116i = 27 + 36i$$

So now find

$\sqrt{27+36i} = a + bi$ for real numbers a and b

$$27 + 36i = (a+bi)^2 = a^2 - b^2 + 2abi$$

Solve

$a^2 - b^2 = 27$ and $36 = 2ab$

simultaneously:

$$b = \frac{18}{a} \text{ so } a^2 - \left(\frac{18}{a}\right)^2 = 27$$
$$a^4 - 324 = 27a^2$$
$$a^4 - 27a^2 - 324 = 0$$
$$(a^2 - 36)(a^2 + 9) = 0$$

Since a is real, $a^2 = 36$ so $a = 6$ or $a = -6$

So that $b = 3$ or $b = -3$

That is $\sqrt{27+36i} = 6 + 3i$ or $-6 - 3i$

$$z = \frac{8 - 5i \pm (6+3i)}{2}$$

$$z = \frac{14}{2} - \frac{2}{2}i \text{ or } z = \frac{2}{2} - \frac{8}{2}i$$

So $z = 7 - i$ and $z = 1 - 4i$

8)

Sum of roots: $a + 6 - 2i = 10 - 2i$ so $a = 4$

Product of roots: $4(6 - 2i) = 24 - 8i$

$\therefore k = 24 - 8i$

9)

Sum of roots: $a + 2 + 7i = k$

Product of roots: $a(2 + 7i) = 6 + 21i$

$$a = \frac{6 + 21i}{2 + 7i} = \frac{3(2 + 7i)}{2 + 7i} = 3$$

So $3 + 2 + 7i = k$

$\therefore k = 5 + 7i$

10)

Since the coefficients are not real, there are no complex conjugate roots.

Sums of roots: $1 + i + 3 + a = 2i \ \ \therefore a = -4 + i$

Sums of pairs of roots: $p = 3(1 + i) + (-4 + i)(1 + i) + 3(-4 + i) = -14 + 3i$

Product of roots: $q = -3(1 + i)(-4 + i) = 15 + 9i$

$\therefore p = -14 + 3i$ and $q = 15 + 9i$

11)

Since the coefficients are not real, there are no complex conjugate roots.

Let a be the remaining root

Sums of pairs of roots: $-4(2 + i) + a(2 + i) - 4a = -5 - 8i$

$$\begin{aligned} -8 - 4i + 2a + ai - 4a &= -5 - 8i \\ a(-2 + i) &= 3 - 4i \\ a &= \frac{3 - 4i}{-2 + i} \times \frac{-2 - i}{-2 - i} = \frac{-10 - 5i}{5} = -2 + i \end{aligned}$$

Sum of roots: $p = -(-4 + (2 + i) + (-2 + i)) = 4 - 2i$

Product of roots: $q = -4(2 + i)(-2 + i) = (-8 - 4i)(-2 + i) = 20$

$\therefore p = 4 - 2i$ and $q = 20$

12)

Since the coefficients are not real, there are no complex conjugate roots.

Let a be the remaining root

Product of roots: $\frac{13}{2} = 3(2 - 3i)a$

$$\begin{aligned} 13 &= (12 - 18i)a \\ a &= \frac{13}{12 - 18i} \times \frac{12 + 18i}{12 + 18i} = \frac{156 + 234i}{468} = \frac{1}{3} + \frac{1}{2}i \end{aligned}$$

Sum of roots: $\frac{p}{2} = -\left(3 + (2 - 3i) + \left(\frac{1}{3} + \frac{1}{2}i\right)\right)$

$$\frac{p}{2} = -\left(\frac{32}{6} - \frac{15}{6}i\right) \Rightarrow p = -\frac{32}{3} + 5i$$

Sums of pairs of roots: $\frac{q}{2} = 3(2-3i) + \left(\frac{1}{3} + \frac{1}{2}i\right)(2-3i) + 3\left(\frac{1}{3} + \frac{1}{2}i\right)$

$$\frac{q}{2} = 6 - 9i + \frac{13}{6} + 1 + \frac{3}{2}i$$
$$\frac{q}{2} = \frac{55}{6} - \frac{15}{2}i \Rightarrow q = \frac{55}{3} - 15i$$

$\therefore p = -\frac{32}{3} + 5i$ and $q = \frac{55}{3} - 15i$

13 a)

Sum of roots: $10 - 2i = a + b + a + bi$

Considering imaginary part: $-2 = b$

Considering real part: $10 = 2a + b$

$\therefore a = 6$ and $b = -2$

13 b)

Sums of pairs of roots: $p = -2(6) - 2(6-2i) + 6(6-2i) = 12 - 8i$

Product of roots: $-q = -2(6)(6-2i) = -72 + 24i$

$\therefore p = 12 - 8i$ and $q = 72 - 24i$

14)

$$6 + 3i = 3 + (1 + 2i) + \alpha$$
$$6 + 3i = 4 + 2i + \alpha$$
$$\therefore 2 + i$$